

Charu Jain

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Professional Summary

14+ years of experience in Artificial Intelligence, Computer Vision, Machine Learning, and Data Engineering. Experienced in designing production AI systems using Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), OCR, Computer Vision, and distributed data processing. Skilled in Python, PyTorch, Spark, Kubernetes, Docker, GCP, AWS, and MLOps with experience deploying scalable enterprise AI applications.

Experience

Data Scientist – Ericsson

Oct 2025 – Present | Noida

- AI-focused experience, pioneering cutting-edge solutions leveraging advanced data engineering.
- Developed a platform for network KPI anomaly detection using statistical methods, reducing anomaly detection time by 40% and enabling proactive fault management across enterprise infrastructure.
- Managed infrastructure for scalable AI services on GCP Kubernetes, automating deployment with Docker and CI/CD pipelines, and optimizing data pipelines using Spark and Postgres.

Senior Module Leader – Motherson Technologies and Services Limited

Jan 2024 – Jul 2025 | Noida

- Engineering Drawing Change Detection using Computer Vision – designed and implemented a system for automated visual comparison of engineering drawing versions, reducing manual inspection by ~500 man-hours/week.
- Crimp cross-section analysis – implemented an image registration pipeline using gradient-based optical flow and UNet-based segmentation, achieving 80% accuracy and 0.82 IoU.
- Led the end-to-end development of a YOLO-based forklift-safety system, directing a cross-functional team to design, containerize, and deploy the solution on Jetson Nano and AWS SageMaker, while establishing monitoring via MLFlow.

Senior Computer Vision Engineer – Vehant Technologies

Dec 2021 – Oct 2023 | Noida

- Under Vehicle Scanning System (UVSS) – conceptualized and prototyped product enhancements, delivering end-to-end module development for production pipelines.
- Solved complex point-cloud registration challenges by implementing a Bayesian coherent point drift algorithm, achieving 95% accuracy and optimizing performance through parallel CPU/GPU processing (CUDA, OpenMP, LAPACK, MAGMA).

API Developer – Barclays

Jan 2018 – Dec 2018 | Pune

- Developed new microservice for the PingIt mobile app using Spring Boot REST framework.
- Deployment and management of microservices using Kubernetes.

Senior Module Lead – Sopra Steria

Jan 2017 – Dec 2017 | Noida

- Developed web application – online quotation tool – using JavaScript, J2EE technologies following Scrum methodology.

Test Engineer (Client: National Australian Bank, American Express) – Infosys

Jul 2011 – Jan 2017 | Melbourne, Mysore

- Internet banking modernization and new business strategy management tool development.
- Internet Banking regression testing.

Projects

Image Segmentation of tooth in CBCT image using iterative 3D U-Net training to build a tooth SSM (Master thesis)

- Automated 3D tooth segmentation from CBCT images using 3D U-Net to create a statistical shape model, evaluating generalisation, specificity and compactness.

3D Slicer extension for editing image labels in 3D

- Developed a user-friendly UI extension using the open-source Slicer framework for editing image labels in 3D.

Combining statistical shape model of complete tooth from CBCT images with laser scan data of its crown (Master project)

- Hybrid 3D model combining CBCT morphology with laser scan details using Metropolis-Hastings and ICP algorithms.

QA bot to read document (Coursera)

- Built a generative AI application using LangChain and Retrieval-Augmented Generation to answer questions from documents.

Generative AI (BossCoder academy)

- Working on an intensive 4-month program on Generative AI and LLM application development, gaining hands-on experience with prompt engineering, embeddings, vector databases, and LLM orchestration using Python, LangChain, ChromaDB, OpenAI API, and Hugging Face models.

Jul 2026 – Oct 2026

Education

University of Basel – Switzerland
M.Sc in Computer Science

Apr 2021

Jaypee University of Information Technology – Solan
B.Tech in Information Technology

Jun 2011

Certifications

ISTQB Certified Tester Foundation Level – ISTQB

Apr 2012

Oracle Certified Professional, Java Programmer – ORACLE

Jul 2011

IBM Generative AI Engineering Professional Certificate – IBM

Aug 2025

Technologies

Programming Languages: Python, Scala, Java, C++, CUDA, OpenMP

Web Development: Angular JS, Springboot (REST, SOAP)

Machine Learning & AI: Statistical shape modelling, Bayesian statistics, Computer vision algorithms, Object detection, Segmentation, Classification, Deep learning, Tensorflow, Keras, LLM, RAG, Langchain